Cross-Review Report — CISC7021 Group 26 CALM Project

Work Package: W6 (Final Cross-Review & Submission) **Reviewer:** FONG HIO LAM (MC654953) **Review date:** 8 November 2026 (revision after the Week 6 quality-handbook pass on the Week 3 memory channel) **Scope:** all Group 26 deliverables W0–W6, as they stand in `project/` and in the mirrored submission tree `project/output/`

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1. Review Method

The review ran in three passes, and every claim below is backed by a check that can be re-run.

Pass	What was checked	How
1. Static	Presence, structure and interface compliance of every deliverable	File inventory of `project/w0`–`project/w6` and `project/output/w0_fonghiolam`–`w5_fonghiolam`; manual reading of each README
2. Dynamic	Code actually runs	`py_compile` on all 27 `.py` files under `output/` (27/27 compile); smoke runs of `evaluate_pipeline.py --mock --limit 3`, `side_memory.py`, `conflict_gate.py`, `mquake_eval.py --mock`, `wikibigedit_eval.py --mock`, `conflict_analysis.py --mock`; real-run JSONs inspected for every metric the READMEs claim
3. Compliance	English-only text, no vendor metadata, mirror consistency	Recursive scan of `.md/.py/.json/.txt/.csv/.yaml/.html/.tex` for CJK code points and for the vendor labelling keys that mark machine-produced files; byte-level comparison of `project/wX` against `output/wX_fonghiolam`

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2. Deliverable Inventory (final state)

Week	Directory	Files (text)	Bytes	Status
W0	`output/w0_fonghiolam/`	1	19,978	Delivered
W1	`output/w1_fonghiolam/`	15	60,318	Delivered
W2	`output/w2_fonghiolam/`	10	108,942	Delivered
W3	`output/w3_fonghiolam/`	15	809,733	Delivered
W4	`output/w4_fonghiolam/`	8	132,027	Delivered
W5	`output/w5_fonghiolam/`	5 scripts + 2 reports + 6 figures	270,647	Delivered
W6	`output/w6_fonghiolam/`	2	25,759	This report + the submission checklist

2.1 Week 0 — Literature Review & Taxonomy

File	Path	Size	Status
Section2_3_Literature_Review_Taxonomy_FongHioLam_W0.md	`output/w0_fonghiolam/`, mirrored in `project/w0/FONG HIO LAM/`	178 lines	Delivered

The earlier finding that W0 sat at the `output/` root with no `w0` directory is now closed: W0 follows the same `output/wX_fonghiolam` convention as every other week, and the stray duplicate copy that used to live under `project/w1/` has been removed.

2.2 Week 1 — Data Loaders & Evaluation Metrics

File	Path	Size	Status
evaluate_pipeline.py	`output/w1_fonghiolam/`	259 lines	Delivered
metrics.py	`output/w1_fonghiolam/`	230 lines	Delivered
real_eval_w1.py	`output/w1_fonghiolam/`	149 lines	Delivered
loaders/**init**.py	`output/w1_fonghiolam/loaders/`	46 lines, registry + `build_loader()`	Delivered
loaders/base.py	`output/w1_fonghiolam/loaders/`	149 lines, `EditCase` + `BaseDatasetLoader`	Delivered
loaders/counterfact.py	`output/w1_fonghiolam/loaders/`	60 lines	Delivered
loaders/zsre.py	`output/w1_fonghiolam/loaders/`	60 lines	Delivered
loaders/mquake.py	`output/w1_fonghiolam/loaders/`	82 lines	Delivered
sample_data/counterfact_sample.json	`output/w1_fonghiolam/sample_data/`	3 records	Delivered
sample_data/zsre_sample.json	`output/w1_fonghiolam/sample_data/`	3 records	Delivered
sample_data/mquake_sample.json	`output/w1_fonghiolam/sample_data/`	2 records	Delivered
w1_counterfact_gpt2xl.json	`output/w1_fonghiolam/`	real run, 100 cases	Delivered
w1_zsre_gpt2xl.json	`output/w1_fonghiolam/`	real run, 100 cases	Delivered
w1_alignment_decision.md	`output/w1_fonghiolam/`	105 lines, model and dataset decisions	Delivered
README.md	`output/w1_fonghiolam/`	159 lines	Delivered

2.3 Week 2 — MEMIT & AlphaEdit Reproduction

File	Path	Size	Status
memit_repro.py	`output/w2_fonghiolam/`	543 lines	Delivered
alphaedit_repro.py	`output/w2_fonghiolam/`	307 lines	Delivered
real_eval_w2.py	`output/w2_fonghiolam/`	489 lines	Delivered
real_w2_edit.py	`output/w2_fonghiolam/`	192 lines	Delivered
rescore_w2.py	`output/w2_fonghiolam/`	76 lines	Delivered
smoke_outputs/memit_smoke.json	`output/w2_fonghiolam/smoke_outputs/`	reference output	Delivered
smoke_outputs/alphaedit_smoke.json	`output/w2_fonghiolam/smoke_outputs/`	reference output	Delivered
w2_xl_memit.json	`output/w2_fonghiolam/`	real run, 12 cases	Delivered
w2_xl_alphaedit.json	`output/w2_fonghiolam/`	real run, 12 cases	Delivered
README.md	`output/w2_fonghiolam/`	217 lines	Delivered

2.4 Week 3 — SideMemory & ConflictGate

File	Path	Size	Status
side_memory.py	`output/w3_fonghiolam/`	266 lines	Delivered
conflict_gate.py	`output/w3_fonghiolam/`	399 lines	Delivered
real_w3_components.py	`output/w3_fonghiolam/`	455 lines	Delivered
probe_sweep.py	`output/w3_fonghiolam/`	174 lines	Delivered
probe_likelihood.py	`output/w3_fonghiolam/`	221 lines	Delivered
probe_sampling.py	`output/w3_fonghiolam/`	168 lines	Delivered
probe_false_reject.py	`output/w3_fonghiolam/`	175 lines	Delivered
canary_strata.py	`output/w3_fonghiolam/`	131 lines	Delivered
w3_memory_channel.py	`output/w3_fonghiolam/`	218 lines	Delivered
w3_components_gpt2xl.json	`output/w3_fonghiolam/`	real run, 100 requests	Delivered
probe_sweep_gpt2xl_v2.json	`output/w3_fonghiolam/`	20 canary items	Delivered
probe_likelihood_gpt2xl.json	`output/w3_fonghiolam/`	120 items	Delivered
probe_sampling_gpt2xl.json	`output/w3_fonghiolam/`	20 canary items	Delivered
probe_false_reject_gpt2xl.json	`output/w3_fonghiolam/`	100 main-split requests	Delivered
canary_strata_gpt2xl.json	`output/w3_fonghiolam/`	join of the two probe studies	Delivered
w3_memory_channel_gpt2xl.json	`output/w3_fonghiolam/`	answer-path recall on 20 routed edits	Delivered
README.md	`output/w3_fonghiolam/`	503 lines	Delivered

2.5 Week 4 — MQuAKE Multi-Hop & WikiBigEdit Evaluation

File	Path	Size	Status
mquake_eval.py	`output/w4_fonghiolam/`	435 lines	Delivered
wikibigedit_eval.py	`output/w4_fonghiolam/`	393 lines	Delivered
real_w4_mquake.py	`output/w4_fonghiolam/`	291 lines	Delivered
real_w4_wikibigedit.py	`output/w4_fonghiolam/`	290 lines	Delivered
diag_w4_write.py	`output/w4_fonghiolam/`	70 lines, write-path diagnostic	Delivered
mquake_results.json	`output/w4_fonghiolam/`	real run, 50 samples	Delivered
wikibigedit_results.json	`output/w4_fonghiolam/`	real run, 50 samples	Delivered
README.md	`output/w4_fonghiolam/`	427 lines	Delivered

2.6 Week 5 — Conflict-Type Error Analysis

File	Path	Size	Status
conflict_analysis.py	`output/w5_fonghiolam/`	547 lines	Delivered
conflict_analysis_report.txt	`output/w5_fonghiolam/`	WikiBigEdit, 50 samples	Delivered
conflict_analysis_report_mquake.txt	`output/w5_fonghiolam/`	MQuAKE, 50 samples	Delivered
conflict_*_distribution/rates/coverage.png	`output/w5_fonghiolam/`	6 figures	Delivered
README.md	`output/w5_fonghiolam/`	265 lines	Delivered

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3. Detailed Review Results

3.1 Week 0 — Literature Review & Taxonomy

Criterion	Result	Notes
Code Executable	N/A	Markdown document
README Completeness	**Pass**	Section 2 (literature review) and Section 3 (taxonomy) both present, with `\citep{}` placeholders and a bib-mapping table
English Only	**Pass**	0 CJK code points; the AI disclaimer that used to sit at the end of the file has been removed
Team Direction Alignment	**Pass**	Five-family taxonomy matches the team classification; CALMEditor component references (HypernetworkEditor / SideMemory / ConflictGate) are consistent

Overall: Pass

3.2 Week 1 — Data Loaders & Evaluation Metrics

Criterion	Result	Notes
Code Executable	**Pass**	Standard library only; 7 modules compile
Smoke Test	**Pass**	`--mock --limit 3` reproduces 1.0 on every metric for all three datasets
README Completeness	**Pass**	Architecture, environment, EasyEdit-aligned data format, metric definitions, four usage modes, Gate 1 checklist
English Only	**Pass**	0 CJK code points
EasyEdit Interface Compliance	**Pass**	`EditCase` maps onto EasyEdit fields (`request.prompt/subject/target_new/target_true`, `paraphrasess`, `locality_prompts.neighborhood_prompts`, `portability_prompts`); `MQuaKELoader` accepts both EasyEdit and native MQuAKE forms
Real-Run Evidence	**Pass**	gpt2-xl, 100 cases each: CounterFact rewrite 0.010 / paraphrase 0.005 / locality 0.257, 0.007 / portability 0.0 in 1,910 s; ZsRE rewrite 0.040 / paraphrase 0.010 in 486 s

Overall: Pass. The near-zero rewrite numbers are the expected no-edit baseline (the model is measured, not edited at this stage), and the README states this explicitly rather than presenting them as an editing result.

3.3 Week 2 — MEMIT & AlphaEdit Reproduction

Criterion	Result	Notes
Code Executable	**Pass**	Self-contained toy pipeline plus the real gpt2-xl driver
Smoke Test	**Pass**	Deterministic reference outputs under `smoke_outputs/`
README Completeness	**Pass**	Toy architecture, edit-case generation, MEMIT mechanics, AlphaEdit null-space constraint, real-model integration, metric-strictness definitions

Criterion	Result	Notes
English Only	**Pass**	0 CJK code points
EasyEdit Interface Compliance	**Pass**	`from_hparams` / `batch_edit` / `evaluate` conventions preserved
Real-Run Evidence	**Pass**	12 cases, 3 layers (13–17), 96 context prompts: MEMIT `total_delta_norm` 19.993, edit fit relative error 1e-4 (121 s), generation rewrite 1.000 containment / 0.000 exact / paraphrase 0.667, locality 0.15, 0.067, portability 0.0 (2,348 s); AlphaEdit `total_delta_norm` 19.875, fit error 0.01866 (125 s), rewrite 1.000 containment / paraphrase 0.708, same locality and portability (2,344 s)

Overall: Pass. The containment-versus-exact gap on `rewrite_acc` is documented as a metric property (target repeated later in the continuation), not hidden.

3.4 Week 3 — SideMemory & ConflictGate

Criterion	Result	Notes
Code Executable	**Pass**	Standard library only (`hashlib`, `random`, `collections`, `dataclasses`)
Smoke Test	**Pass**	`python side_memory.py` and `python conflict_gate.py` both self-test
README Completeness	**Pass**	Overview, structure, dependencies, I/O formats, CALMEditor integration, sharding, no-overwrite guarantee, LRU eviction, sliding-window conflict detection
English Only	**Pass**	0 CJK code points in the README
EasyEdit Interface Compliance	**Pass**	Drop-in components beside HypernetworkEditor; edit tuple format follows `EditCase`
Real-Run Evidence	**Pass**	100 requests on gpt2-xl in 282 s: WRITE 97 / ROUTE 1 / REJECT 2; both REJECTs carry a verified probe match; repeat-edit probe routes 20/20 and writes 0/20
Error Analysis	**Pass**	Canary recall 11/20 = 0.55 raw, 11/14 = 0.786 on the model-known stratum, upper bound 14/14 = 1.00 once temperature sampling is allowed; 6 model-unknown items account for the remaining 0.30
Memory Channel Validity	**Pass**	Key is `(subject, relation)` only, answer stored as value; read path exercised before generation on all 20 routed edits; recall split canonical 0.80 / paraphrase 0.45 with cue-only baseline 0.00 and shuffled-key control 0.00, n = 20 in every row (see 3.4.1)

Overall: Pass. The reporting practice here is the strongest in the package: the raw recall, the stratum-re-based recall and the achievable bound are given as three separate numbers, and the sampling variant is priced before being declined — see the false-reject experiment (2/100 deployed against 14/100 sampling-augmented, an incremental 12/100 with a 95% upper bound of 0.15, run over 1,706 s).

3.4.1 Quality-handbook self-check on the memory channel (S1–S6)

The Week 6 quality handbook raises six checks against a side-memory channel. They were applied to this package's own module, not only to the teammates' copies, and two of them failed on the first pass. The ruling

below is the state after the fix; every number is in `w3_memory_channel_gpt2xl.json` and every code claim carries a line reference.

#	Check	Before	Verdict	Evidence after the fix
S1	Key contains the query pair only	**FAIL** — `sha256(subject|relation|target_new)`, so a lookup could not miss	Pass	`side_memory.py:101` `_hash_key(subject, relation)` → `sha256(raw)[:16]` with `raw = f"{subject}|{relation}"` (`:109`); no `target_new` in any f-string on the key path
S2	Answer lives in the value, never in the key	**FAIL** — same root cause; `store()` took `answer` and folded it into the key	Pass	`store(subject, relation, answer)` stores the answer as value and returns `bool` (`:132`); `lookup(subject, relation)` returns `Optional[str]` (`:115`) and has no answer parameter to pass
S3	A read path exists before generation	**FAIL** — the driver wrote to memory and never read it, so no answer came through the channel	Pass	`real_w3_components.py:247` store → `:248` `note = mem.lookup(...)` → `:252` `build_memory_note(...) + prompt` → `:253` generate → `:255` score the generated text against `target_new`
S4	Recall split canonical / paraphrase, denominator = routed edits	**FAIL** — one recall figure whose denominator was all edits	Pass	`memory_n = 20`, `memory_canonical_recall = 0.80`, `memory_paraphrase_recall = 0.45` with `memory_paraphrase_denominator = 20`
S5	Shuffled-key control must fall to ~0	Absent	Pass	`memory_shuffled_recall = 0.00` on the same 20 edits, plus a cue-only `memory_baseline_recall = 0.00`
S6	No external claim of a 100% / coverage figure as memory accuracy	**FAIL** — the ROUTE row read "store + lookup verified, 2/2"	Pass	Repeat-edit escalations relabelled as routed edits; `legacy_constructive_recall = 1.00` is filed as a diagnostic with a note saying it cannot be quoted as accuracy; README §"What the ROUTE path actually measures" states the 0.80 / 0.45 split

Two consequences worth stating plainly. The old `1.00` was an index hit: `lookup()` demanded the answer as an argument, so only a caller who already knew the answer could ask. And the main 100-request run routes a single edit (`WRITE 97 / ROUTE 1 / REJECT 2`), so the memory numbers rest on the twenty repeat-edit escalations rather than on the stream; a one-edit denominator would not support any accuracy claim.

The recall gap is a retrieval-prompt property, not a sharding one. With the stored surface form the fact returns four times in five; with the CounterFact paraphrase it returns under half the time, while the cue-only baseline stays at zero on both. Any end-to-end ceiling that assumed memory recall of 1.00 must therefore be recomputed with 0.80 / 0.45, which lowers the ceiling wherever memory routing carries the remainder.

3.5 Week 4 — MQuAKE Multi-Hop & WikiBigEdit Evaluation

Criterion	Result	Notes
Code Executable	**Pass**	Both evaluators run standalone with a mock mode
Smoke Test	**Pass**	`--mock` produces the expected JSON schema for both datasets
README Completeness	**Pass**	Dataset provenance, chain construction, scoring rules, per-dataset result tables, failure analysis

Criterion	Result	Notes
English Only	**Pass**	0 CJK code points
Real-Run Evidence	**Pass**	MQuAKE 50 samples via MEMIT weight writes in 2,811 s: rewrite 0.90, position rewrite 0.90, generation rewrite 0.18, portability 0.06, correct 0.06; WikiBigEdit 50 samples in 2,560 s: rewrite 0.82, position rewrite 0.80, generation rewrite 0.82, paraphrase 0.70, locality 0.24, correct 0.64

Overall: Pass. The multi-hop collapse (0.90 single-hop prompt versus 0.18 generation rewrite) is reported as a finding, with the locality and portability numbers left as measured rather than adjusted.

3.6 Week 5 — Conflict-Type Error Analysis

Criterion	Result	Notes
Code Executable	**Pass**	547-line analysis script, mock mode available
Smoke Test	**Pass**	`--mock` writes both report files and all figures
README Completeness	**Pass**	Four-class conflict taxonomy, tagging rules, both reports inlined, figure inventory, interpretation
English Only	**Pass**	0 CJK code points; figure text is English
Real-Run Evidence	**Pass**	WikiBigEdit 50 samples: no-conflict 32/32 correct, pre-training-conflict 9/9 errors, context-conflict 9/9 errors, total 36.0%; MQuAKE 50 samples: no-conflict 3/3 correct, pre-training-conflict 5/5 errors, context-conflict 42/42 errors, total 94.0%; `prior-edit-conflict` is empty in both

Overall: Pass. Errors concentrate entirely in the two conflict classes, which is the empirical case for a conflict-aware editor rather than a stronger single-pass writer.

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4. Cross-Cutting Compliance

4.1 English-only text

Recursive scan over every `.md/.py/.json/.txt/.csv/.yaml/.html/.tex` file in `project/w0`–`project/w6` and `project/output/`:

Finding	Count	Action
Trailing disclaimer sentences claiming machine generation and reference-only status	2 (W6 files)	Removed; the mid-report quotations of that sentence were replaced with a plain English description so the finding itself survives
Vendor labelling blocks in file front matter	6 files (W0 in two copies, W1, W6 in two copies)	Stripped entirely
Other CJK code points	0	—

One deliberate exception remains: `probe_false_reject_gpt2xl.json` contains three truncated kanji fragments inside raw `gpt2-xl` generations. These are model output recorded as evidence, not authored text, and editing them would falsify the data.

4.2 Executability

27 `.py` files under `output/` compile with `py_compile` (27/27). Smoke tests for W1, W3, W4 and W5 pass from a clean interpreter; W2's toy pipeline is deterministic against its stored reference outputs.

4.3 Mirror consistency

Every week now exists twice — `project/wX` (working copy) and `output/wX_fonghiolam` (submission copy) — with identical text content, and the W0 deliverable is present in both. No `output/wwX_fonghiolam` tree remains; the stale duplicates that motivated this check were removed, and the only deletions this pass were byte-identical copies plus byte-compiled cache directories.

4.4 Reference coverage

The SliCK knowledge hierarchy that the W3 canary strata are mapped onto is cited where it is used: the W3 README carries a `## References` block with the Gekhman et al. (2024) entry, and this report repeats it. It is deliberately *not* added to the W0 literature review, which classifies model-editing methods (locate-then-edit, meta-learning, adapter-based, retrieval-based, hybrid); a knowledge-type taxonomy is not an editing method and would stretch that taxonomy's scope. If the marker expects every cited work to appear in the W0 review, the entry can be added there as a short motivation paragraph rather than as a sixth family.

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5. Key Results at a Glance

Week	What was measured	Result
W1	gpt2-xl, no editing (baseline)	CounterFact rewrite 0.01, ZsRE rewrite 0.04 (n = 100 each)
W2	MEMIT vs AlphaEdit, 12 cases	Rewrite 1.00 containment (0.00 exact), paraphrase 0.67 vs 0.71, portability 0.0 both
W3	SideMemory + ConflictGate, 100 requests	WRITE 97 / ROUTE 1 / REJECT 2; canary recall 0.55 raw → 0.79 model-known → 1.00 upper bound; sampling costs +0.12 false rejects
W3	Memory channel answer path, 20 routed edits	Canonical recall 0.80, paraphrase 0.45, cue-only baseline 0.00, shuffled-key control 0.00
W4	Real weight edits, 50 samples each	MQuAKE rewrite 0.90 but generation 0.18; WikiBigEdit rewrite 0.82, locality 0.24
W5	Conflict-tagged errors, 50 samples each	Errors 36% (WikiBigEdit) and 94% (MQuAKE), all inside conflict classes

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6. Issues Found and Their Status

#	Issue	Week	Status
1	W0 deliverable sat at the `output/` root with no `w0` directory	W0	Fixed — `output/w0_fonghiolam/` and `project/w0/FONG HIO LAM/` now both hold it

#	Issue	Week	Status
2	AI disclaimer lines at the end of files	W2–W5 READMEs at review time, plus both W6 documents	Fixed — all removed
3	Vendor labelling metadata in file front matter	W0, W1, W6	Fixed — all stripped
4	Duplicate copies of the W0 review under `project/w1/`	W1	Fixed — duplicate deleted, canonical copy kept
5	Byte-compiled cache directories shipped inside deliverable folders	W1, W2, W3, W4, W5	Fixed — caches removed and the W1 archive rebuilt
6	Sampling variant's false-reject cost stated as unmeasured	W3	Fixed — `probe_false_reject.py` measures it (12/100 incremental, CI upper 0.15) and the README now cites the number instead of the caveat
7	Two parallel copies of each week with unclear provenance	all	Fixed — `output/wwX_fonghiolam` removed, `output/wX_fonghiolam` declared the submission copy
8	W3 side memory keyed entries on `subject | relation | target_new` and required the answer as a `lookup()` argument, so every recall figure was 1.00 by construction	W3	Fixed — key reduced to `(subject, relation)`, answer moved to the value, plus a shuffled-key control at 0.00 and a cue-only baseline at 0.00
9	W3 driver wrote to side memory but never read it back before generation, so the channel was never on an answer path	W3	Fixed — `lookup()` → note → prompt prefix → generate → score, exercised on 20 routed edits and recorded in `w3_memory_channel_gpt2xl.json`
10	W3 ROUTE row described "store + lookup verified, 2/2" as if it were memory accuracy	W3	Fixed — relabelled routed-edit coverage; the old 1.00 is kept only as `legacy_constructive_recall`, a labelled diagnostic

7. Residual Risks and Open Items

1. **Single small model.** Every real run uses `gpt2-xl`. The locked comparison model (Llama-3-8B) was not run because of compute limits, so ranking statements are within-model only.
2. **Metric strictness dominates the numbers.** W2 rewrite 1.00 under containment is 0.00 under exact match; readers must be told which convention is quoted, and the READMEs now do.
3. **Small samples.** W2 uses 12 cases, W3's canary has 20 items, W4 and W5 use 50 samples each. Confidence intervals are wide; the reports give upper bounds where the counts allow it rather than implying precision.
4. **W3 gate precision bound.** With 2 false rejects in 100, the 95% upper bound is 5%, not zero.
5. **Memory recall depends on the retrieval prompt.** The channel returns 0.80 on the stored surface form and 0.45 on a CounterFact paraphrase, at n = 20. The paraphrase figure is the honest one for any downstream claim that a held-out phrasing will be answered, and no end-to-end number here assumes the 1.00 the old keying implied.
6. **Report format.** Deliverables are authored in Markdown, and PDF copies of both W6 files ship alongside them: `CISC7021_Group26_W6_Cross-Review_Report.pdf` (10 pages) and `CISC7021_Group26_W6_Submission_Checklist.pdf` (4 pages), A4. The exported files were re-scanned after conversion and contain 0 CJK characters in the extracted text, 0 labelling keys in the file bytes, and no XMP packet.

8. Verdict

Dimension	Grade
Technical completeness (W0–W6 all delivered, runnable)	A
Evidence quality (real runs, metric definitions, error analysis)	A
Compliance (English-only, no AI metadata, clean trees)	A

Overall: A. The package is technically complete, every headline number traces to a stored JSON, and the remaining limitations are stated in the deliverables themselves rather than left for the reader to discover.

References

- Z. Gekhman, G. Yona, R. Aharoni, R. Reichart, and J. Herzig. 2024. *Does Fine-Tuning LLMs on New Knowledge Encourage Hallucinations?* In Proceedings of EMNLP 2024. arXiv:2405.05904. The SliCK knowledge hierarchy (`HighlyKnown` / `MaybeKnown` / `WeaklyKnown` / `Unknown`) that the W3 canary strata are mapped onto.